Phase 0 — FCS-coordinate reproduction of Property 5

`deq/closed_form_neg_loop/`, May 2026

1. Goal

Reproduce De Maria et al. 2020 §6.2.5, **Property 5** (oscillation in a negative loop): given two delayer-style neurons in a negative-loop motif and a constant input sequence of 1s, the activator A fires the periodic pattern 1100 and the inhibitor I echoes one tick later. Run the FCS-accurate oracle `deq/archetypes/lif_fcs.py:simulate` verbatim over a 2-D scaled-integer grid in (w_{IA}, w_{XA}), with $w_{AI} = 11$ fixed.

2. Setup

- **Oracle**: `lif_fcs.simulate` with `tau = 105`, `rvector = [10, 5, 3, 2, 1]`.
- **Topology**: `topologies.negative_loop(w_XA, w_AI, w_IA)`. Two neurons A (activator, index 0) and I (inhibitor, index 1). A receives external X (weight w_{XA}) and inhibitory feedback from I (weight $w_{IA} < 0$). I is excited by A (weight $w_{AI} > 0$).
- **Grid**: $(w_{IA}, w_{XA}) \in \{-40, ..., -1\} \times \{1, ..., 40\}$, $40 \times 40 = 1600$ cells. w_{AI} fixed at 11 (FCS default).
- **Stimulus**: constant external input $X(t) = 1 \forall t$.
- **Window**: $T_{\max} = 64$ ticks, post-warmup window $t \in [16, 63]$ (48 ticks = 12 candidate periods).

Two per-cell labels:

- **strict_p5**: A's post-warmup spike train matches some cyclic rotation of `1100`, repeated.
- **broad_osc**: A's post-warmup train has a regular period $p \in [2, 12]$ with mixed firing (at least one spike and one silence per cycle).

3. Result

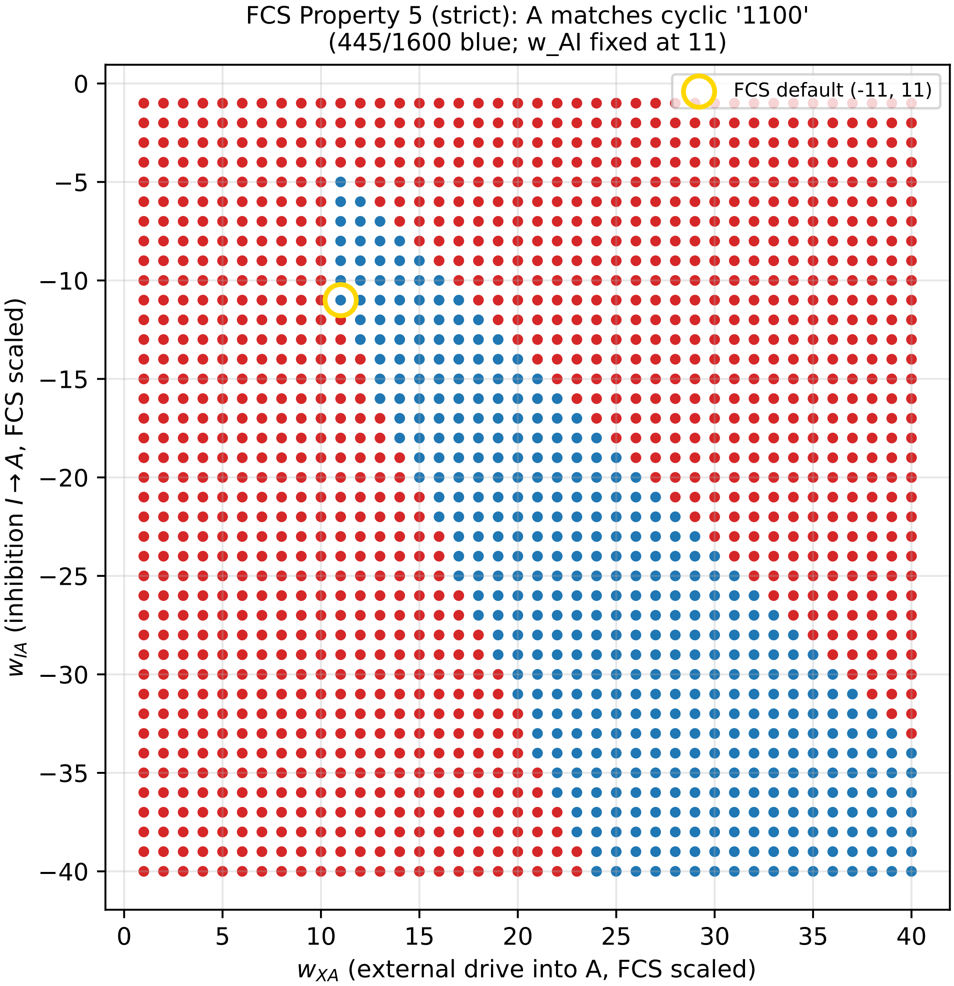

Figure 1: Strict Property 5: cells whose activator spike train is a cyclic rotation of `1100`. Gold ring marks the FCS default cell $(w_{IA}, w_{XA}) = (-11, 11)$ where Property 5 is canonically stated. **445 / 1600 = 27.8%** of cells satisfy the strict pattern.

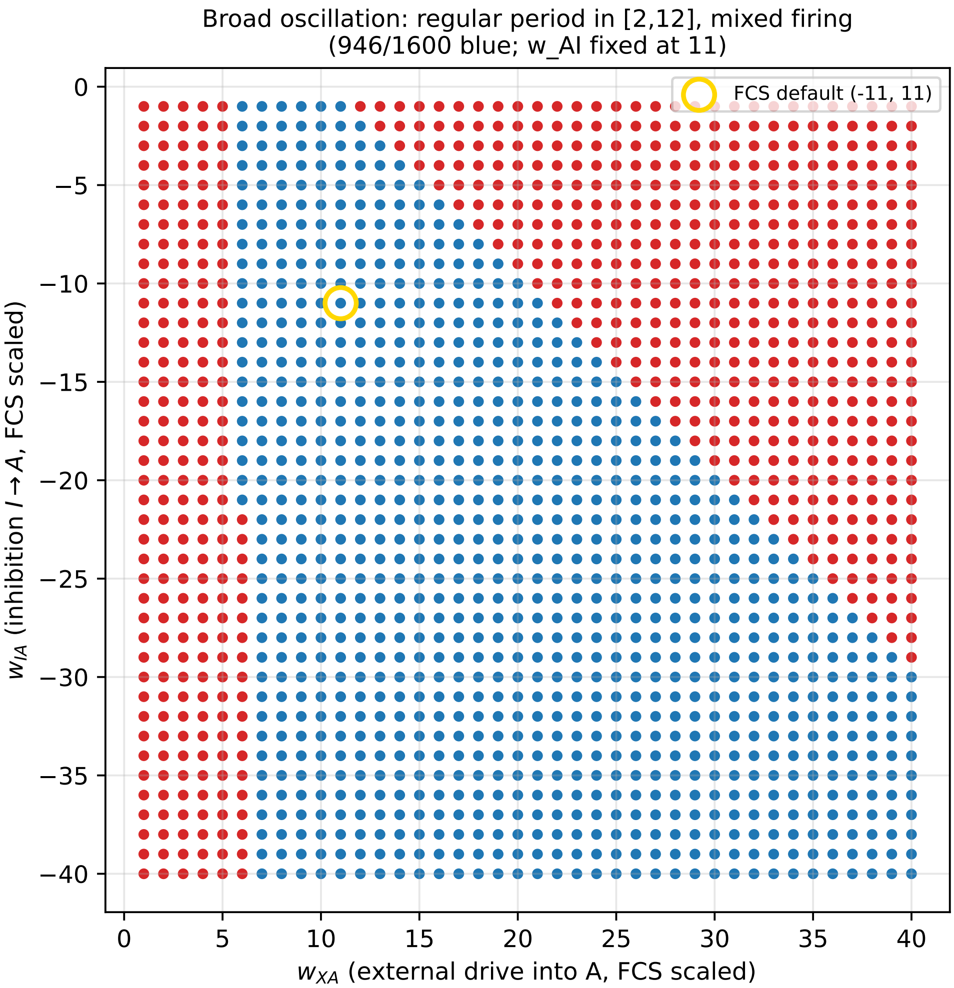

Figure 2: Broad oscillation: cells with any regular period $\in [2, 12]$ and mixed firing. **946 / 1600 = 59.1%**. The complement consists of saturation cells (period 1, A fires every tick — when external drive overwhelms inhibition) and silent cells (drive too weak to push A above threshold).

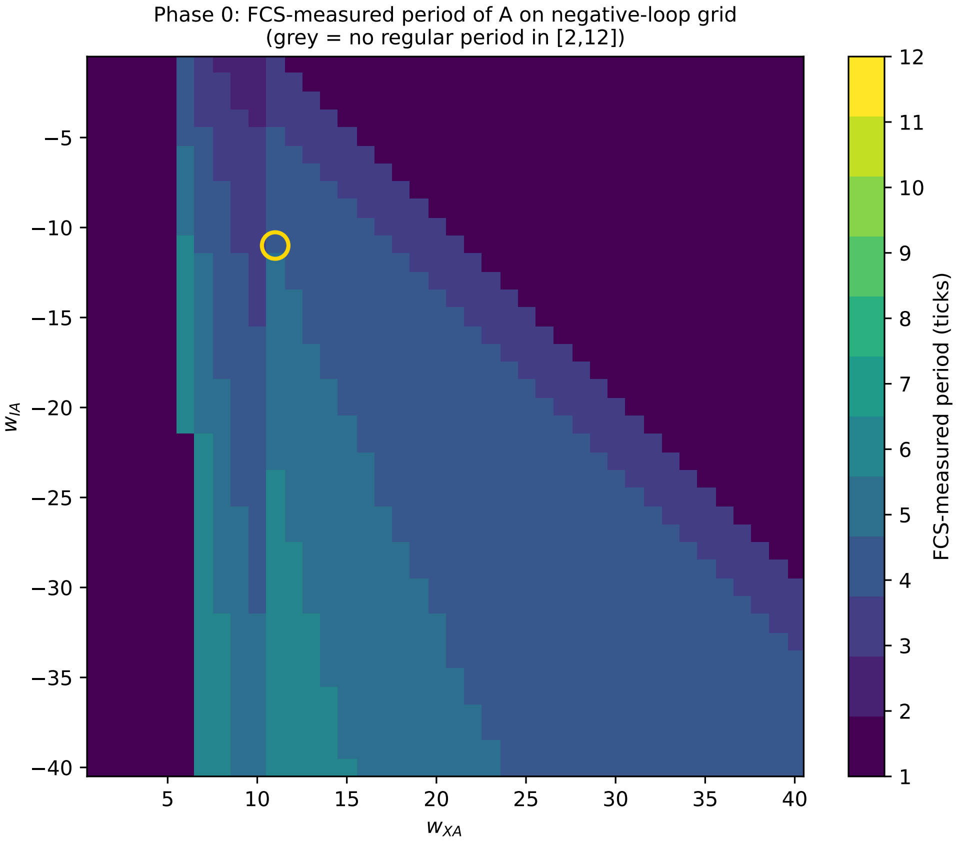

Figure 3: FCS-measured period of A (grey = no regular period in $[2, 12]$, mostly saturation at $w_{XA} \geq |w_{IA}|$ where A fires every tick). The period-4 band (yellow-green) hugs the $w_{XA} \sim |w_{IA}|$ diagonal — the regime where the external drive is approximately balanced by the inhibitory feedback. Period 3, 5, 6 bands flank it.

Period	Cells	Reading
1	654	A fires every tick (drive overwhelms inhibition)
2	8	co-fire / period-2 lock
3	149	period-3 lock
4	498	Property-5 territory (1100 family)
5	207	longer cycles
6	84	longer cycles

4. Sanity gate

FCS default cell $(w_{IA}, w_{XA}) = (-11, 11)$ reproduces Property 5 verbatim:

$$A : 0110011001100110011001100110\ldots \quad I : 0011001100110011001100110011\ldots$$

so strict_p5 $= 1$, period $= 4$. **Phase 0 PASS.**

5. Verdict

The negative-loop FCS oracle exhibits a coherent period-4 region ($w_{XA} \approx |w_{IA}|$, the "balanced" diagonal) where Property 5 holds, flanked by other periodic regions (3, 5, 6) and by saturation regions (period 1) where the inhibition is too weak to interrupt A. Phase 1 will check whether Siegert rate theory predicts these regions from fixed-point + Jacobian structure; Phase 2 will check whether the $H(\omega)$ ringing-period prediction $T_{\text{pred}} = 2\pi/\left|\text{Im}(\lambda)\right|$ lands at 4 on the period-4 band; Phase 3 will check whether quasi-renewal noise at finite N sustains the ringing into a stochastic limit cycle.

Output: `results/phase0/{fcs_grid.npz, prop5_strict.pdf, osc_broad.pdf, period_map.pdf}`, `results/phase0.log`.